- **Pick one from the following:**
  - DSCI 530: Statistical Methods
  - DSCI 575: Advanced Machine Learning
  - CSCI 576: Deep Learning
  - CSCI 578: Bioinformatics (Assignments are geared towards analytics *)
  - CSCI 507: Computer Vision (Assignments are geared towards analytics *)

** Note: The assignments are focused on applying analytical techniques and methodologies, with an emphasis on data analysis, interpretation, and deriving insights from data. They are designed to develop students' skills in exploring, processing, and making data-driven decisions using various tools and frameworks commonly used in analytics.*

Track 2: Cybersecurity

The track in Cybersecurity is targeted to train recent graduates or mid-career professionals with a bachelor's degree in computer science or a related field with a foundational knowledge in programming, data structures, algorithms, systems programming, statistics, and discrete math.

To complete the track in Cybersecurity, students must complete 9 hours of coursework as follows:

- **Required courses:**
  - CSCI 574: Theory of Cryptography
  - CSCI 585: Information Security and Privacy
- **Pick one from the following:**
  - CSCI 560: Fundamentals of Computer Networks
  - CSCI 587: Cyber Physical Systems Security

Track 3: Analytics and Project Management

The track in Analytics and Project Management is targeted to train recent graduates or mid-career professionals with a bachelor's degree in computer science or other STEM-related field with acquired basic knowledge in programming, data structures, statistics, and linear algebra.

To complete the track in Analytics and Project Management, students must complete 9 hours of coursework as follows

- **Required course:**
  - EBGN 525: Business Analytics
- **Pick two from the following:**
  - EBGN553: Project Management
  - EBGN 563: Management of Technology and Innovation (taught by Scott Houser)
  - EBGN 565: Marketing for Technology-based companies
  - EBGN 571: Marketing Analytics
  - EBGN 577: Leading and Managing High Performance Teams

Projects in these courses would focus on software development or data science/machine learning, with an emphasis on requiring more in-depth software development and analysis. As the online program expands, we plan to add more computer science-oriented courses to this certificate track, such as cybersecurity leadership and operations.

Track 4: #Advanced Software Technologies

The track in Advanced Software Technologies is targeted to train recent graduates or mid-career professionals with a bachelor's degree in computer science or other STEM-related field with acquired basic knowledge in programming, data structures, and software engineering.

To complete the track in Advanced Software Technologies, students must complete 9 hours of coursework as follows:

- **Required courses:**
  - CSCI 520: Software Engineering Systems Development I
  - CSCI 521: Software Engineering Systems Development II
- **Pick one from the following:**
  - CSCI 578: Bioinformatics (Assignments are geared towards software design ^)
  - CSCI 507: Computer Vision (Assignments are geared towards software design ^)

^ Note: The assignments are focused on developing and applying software design principles, emphasizing the creation of robust, scalable, and efficient systems. They are designed to enhance students' skills in structuring, designing, and implementing software solutions, while considering best practices, design patterns, and user needs within real-world scenarios. k

Graduate Certificate in Cybersecurity

The online Graduate Certificate in Cybersecurity is targeted to train recent graduates or mid-career professionals with a bachelor's degree in computer science or a related field with a foundational knowledge in programming, data structures, algorithms, systems programming, statistics, and discrete math.

To complete the online Graduate Certificate in Cybersecurity, students must complete 9 hours of coursework as follows:

- Required Courses:
  - o CSCI 574: Theory of Cryptography
  - o CSCI 585: Information Security and Privacy
- Pick one from the following:
  - o CSCI 560: Fundamentals of Computer Networks

- CSCI 587: Cyber Physical Systems Security

While all courses are to be delivered online, students that are interested in attending an on-campus class may substitute one of the above courses with a pre-approved elective as listed in our Graduate Catalog.

FUTURE PLANS

In the 2025-2026 academic year, we aim to expand our course offerings in collaboration with our external advisory committee to meet the evolving needs of working professionals. Based on our ongoing conversations, we plan to introduce a new course focused on Cybersecurity Leadership and Operations to address key industry priorities. This course, designed to support students in preparing for the CISSP exam from ISC2, will complement the following curriculum:

- Required Courses:
  - CSCI 574: Theory of Cryptography
  - CSCI 585: Information Security and Privacy
- Pick one from the following:
  - CSCI 560: Fundamentals of Computer Networks
  - CSCI 587: Cyber Physical Systems Security
  - CSCI 5xx: Cybersecurity Leadership and Operations (this course is added to help prepare students for the CISSP exam from ISC2).

Courses

CSCI500. GRADUATE SEMINAR. 0.0 Semester Hrs.
This zero-credit graduate course builds on the CS department seminars in the colloquium series, which consist of presentations delivered by external or internal invited speakers on topics broadly related to computer science. The seminar is mandatory for all Ph.D. students. The students are required to enroll in the course every semester. Any student who cannot take the course for valid reasons should notify their advisor, who will then make a request to the CS graduate committee for a waiver. These requests could be for the duration of one semester or longer. The course will be graded as PRG/PRU based on student attendance at the department seminars in the colloquium series - the student has to attend at least two thirds of all the seminars each semester in order to get a PRG grade.
Course Learning Outcomes

CSCI507. INTRODUCTION TO COMPUTER VISION. 3.0 Semester Hrs.
Equivalent with CSCI437,CSCI512,EENG507,EENG512,
(I) Computer vision is the process of using computers to acquire images, transform images, and extract symbolic descriptions from images. This course provides an introduction to this field, covering topics in image formation, feature extraction, location estimation, and object recognition. Design ability and hands-on projects will be emphasized, using popular software tools. The course will be of interest both to those who want to learn more about the subject and to those who just want to use computer imaging techniques. 3 hours lecture; 3 semester hours. Prerequisite: Undergraduate level knowledge of linear algebra, statistics, and a programming language.
Course Learning Outcomes

- 1. Be able to analyze and predict the behavior of image formation, transformation, and recognition algorithms
- 2. Be able to design, develop, and evaluate algorithms for specific applications
- 3. Be able to use software tools to implement computer vision algorithms
- 4. Communicate (in oral and written form) methods and results to a technical audience

CSCI508. ADVANCED TOPICS IN PERCEPTION AND COMPUTER VISION. 3.0 Semester Hrs.
Equivalent with EENG508,
(II) This course covers advanced topics in perception and computer vision, emphasizing research advances in the field. The course focuses on structure and motion estimation, general object detection and recognition, and tracking. Projects will be emphasized, using popular software tools. Prerequisites: EENG507 or CSCI507. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- 1. Be able to review the literature on computer vision and create a critical review
- 2. Be able to design, develop, and evaluate algorithms for specific applications
- 3. Be able to use software tools to implement computer vision algorithms
- 4. Communicate (in oral and written form) methods and results to a technical audience

CSCI519. SOFTWARE DESIGN PATTERNS. 3.0 Semester Hrs.
This course focuses on fundamental software design patterns for object-oriented design to create extensible and maintainable software solutions. Design patterns provide time tested solution templates to recurring architectural challenges in com ex software systems. Core topics include Gang of Four (GoF) design patterns, antipatterns, and refactoring to patterns, where students apply multiple patterns in conjunction to modify non-toy code examples. The Model-View-Controller (MVC) architecture is examined as a compound pattern to dissect how individual patterns interact within a larger system. Lectures are supplemented by in-class exercises focused on UML modeling or code implementation. Assignments require students to solve specific scenarios by providing a design justification, UML diagrams, and implementation, all of which are evaluated against SOLID principles. By the end of the course, students will be able to efficiently communicate program structures and develop high-quality, professional software. Prerequisites: CSCI306.
Course Learning Outcomes

- Create a moderately complex software application that integrates one or more software design patterns to ensure reusability, maintainability, and adherence to object-oriented design principles.
- Evaluate an object-oriented software design based on its flexibility, extensibility, and maintainability, using SOLID principles and object-oriented best practices.
- Transfer design principles derived from patterns to refactor existing object-oriented systems and to develop high-quality solutions for new software design scenarios.
- Analyze a software design problem to select and justify an appropriate design based on the problem context and design goals.
- Analyze object-oriented software designs to identify antipatterns and their impact on system quality and maintainability.